

HW 1.5b

Name(s): _____

1. A Tetherball is hit so that it takes 3 seconds to spin around the central pole one time. The mass of the Tetherball is 0.5 kg. The length of the string is 1.2 m.

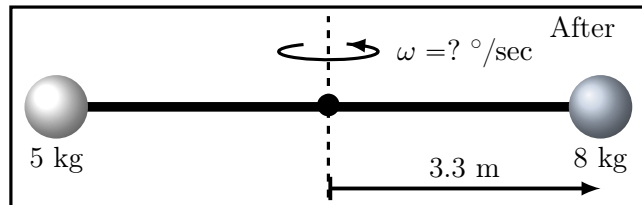
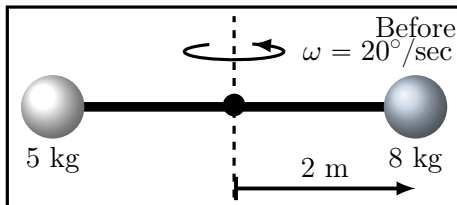
A. Draw a picture of this scenario

B. Determine the spin rate ω , of the Tetherball in units of degrees per second

C. Determine the moment of Inertia of the Tetherball, I

D. Calculate the angular momentum of the Tetherball, L

2. A dumbbell made out of a very light rod attached to two masses (made from two different metals) spins as shown. The masses are each moved further away from the central spin axis. Find the final spin rate (ω), in units of degrees per second.



3. Explain why when a spinning ice skater unfolds their arms their spin rate slows

4. Jupiter's Mass is 318 times that of the Earth. It orbits at a distance of 5.2 AU from the Sun and has an orbital period, $P = 11.86$ Earth years.

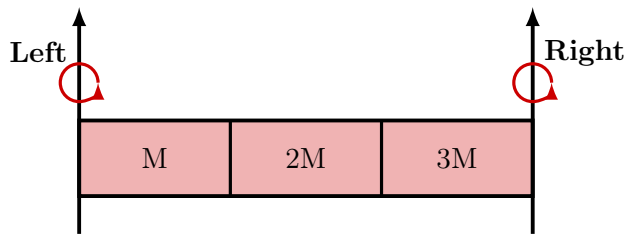
A. Use Jupiter's orbital period to calculate the spin rate of Jupiter, in degrees per second. *This means that you need to find the orbital period in seconds*

B. Calculate Jupiter's moment of Inertia, in units of kg m^2 . *This means that you will need to use kilograms for Jupiter's mass, and meters for its orbital radius.*

C. Calculate the Angular momentum of Jupiter

D. If Jupiter were somehow to magically gain mass, what would happen?

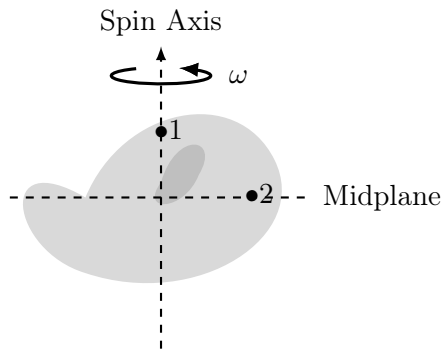
5. Is it harder to spin the rod around its left end or right end?



6. *Hydrostatic Equilibrium* is a balance in any object between the inward pull of _____ and the outwards push of _____.

7. Explain the nebular hypothesis for the origin of the solar system. Use the words giant molecular cloud, hydrostatic equilibrium, conservation of angular momentum.

8. An image of a **Giant Molecular cloud** is shown. If it were to cool as it slowly radiates its energy away, what would happen to _____.



A. The thermal pressure in the space cloud?

B. The force of gravity in the space cloud?

C. The size of the space cloud?

D. The spin rate?

E. The location of particle 1 ?

F. The location of particle 2 ?

G. Draw the resulting *Protoplanetary disk* that would result

9. All the planets in the solar system revolve around the Sun in the same direction. This implies the planets formed _____.

A. at different times and different places and then fell into orbit around the Sun

B. from a single large cloud of gas and dust that was rotating in the same direction as the planets now revolve

C. from gas and dust that was rotating opposite to the current direction of planetary orbit around the Sun

10. Imagine that the planets did not form within the solar system, but instead were accreted from the empty space surrounding it as isolated objects joined the Sun. What sort of shape would you expect the planets' orbits to form?
11. How many of the nine planets revolve around the Sun in a counter-clockwise direction (as viewed from above the North Pole)?
12. Why is the solar system shaped like a disk, rather than a sphere?
13. Explain how the protoplanetary disk eventually evolved into nine planets orbiting around a central star.